Di□usion models

Forward and backward process, generation, training, conditioning

Carlos Cotrini

CAS Building ML/AI Applications, ETH Zurich, weekend 4

Friday 23 October 2026

Which are real? Which are fake?



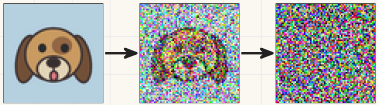
Di□usion models in brief

A di□usion model generates new data, such as images, by learning to undo a process that gradually adds noise.

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Training

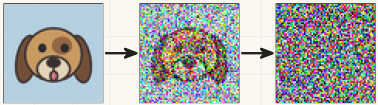


Take real images, add noise in many small steps, and learn to predict the noise that was added.

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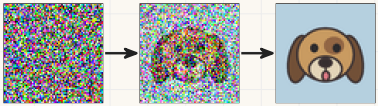
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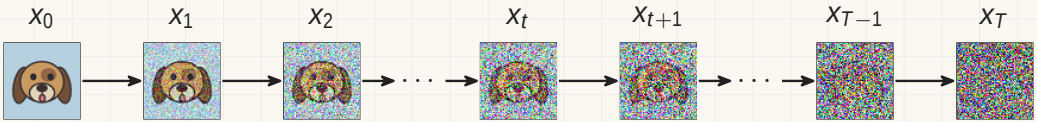
Generation



Start from pure noise and remove the predicted noise, step by step, until a clean image appears.

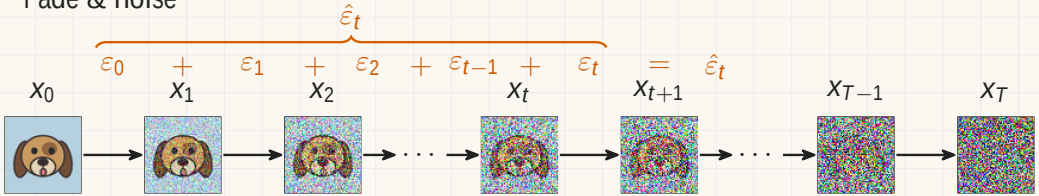
Forward & backward process

Fade & noise

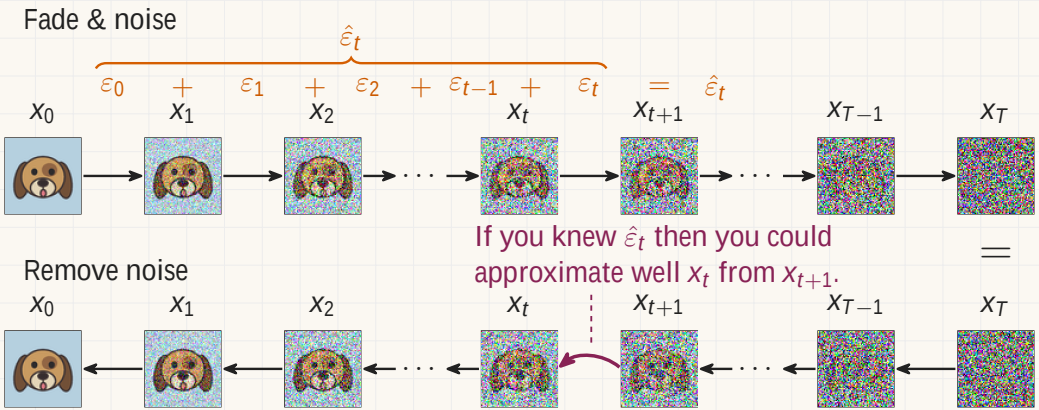


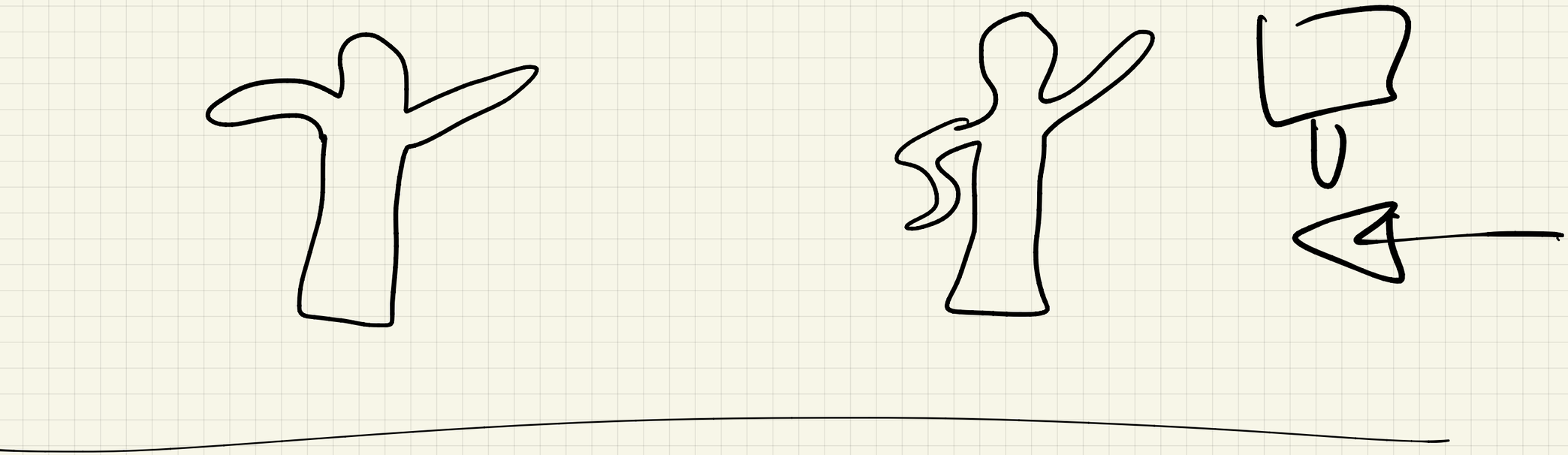
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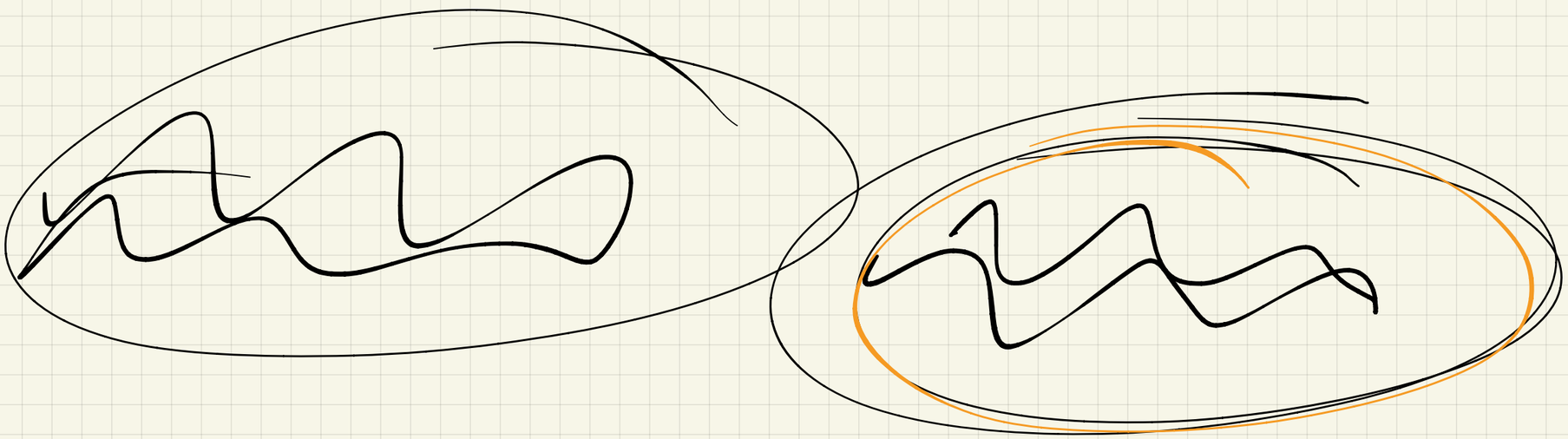
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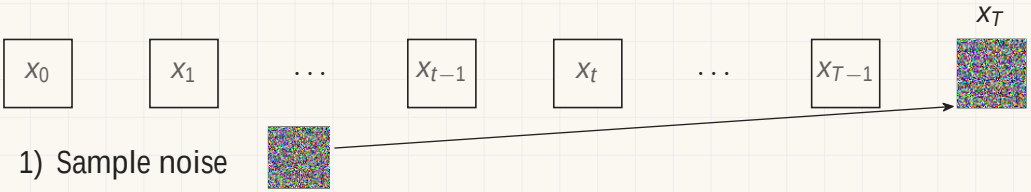
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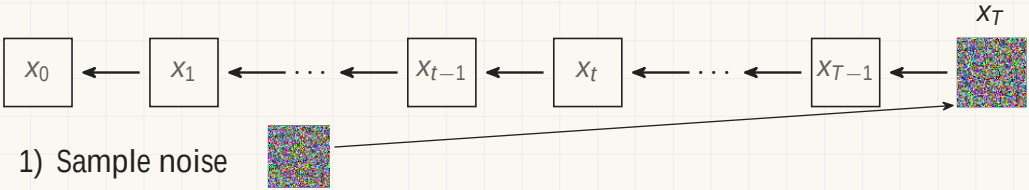




Generating from di□usion

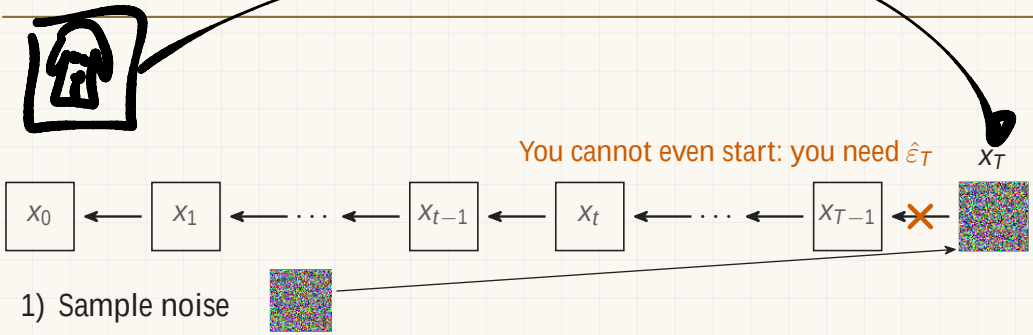


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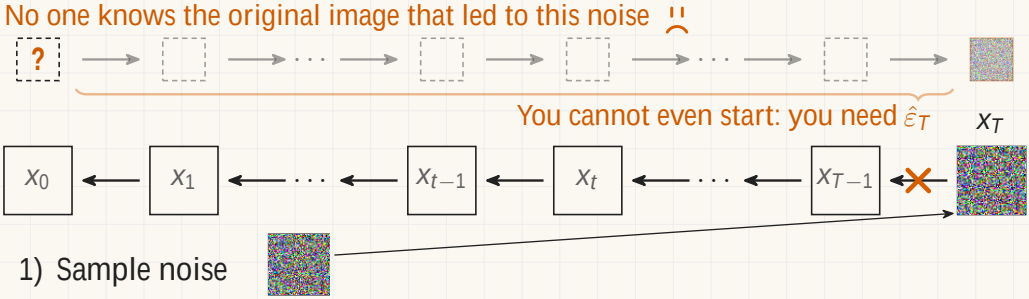
1) Sample noise
2) For $t = T, T-1, T-2, \ldots, 1$:
3) Output x_0

Generating from di□usion



1) Sample noise
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   Estimate x_{t-1} from x_t using $\hat{\varepsilon}_t$
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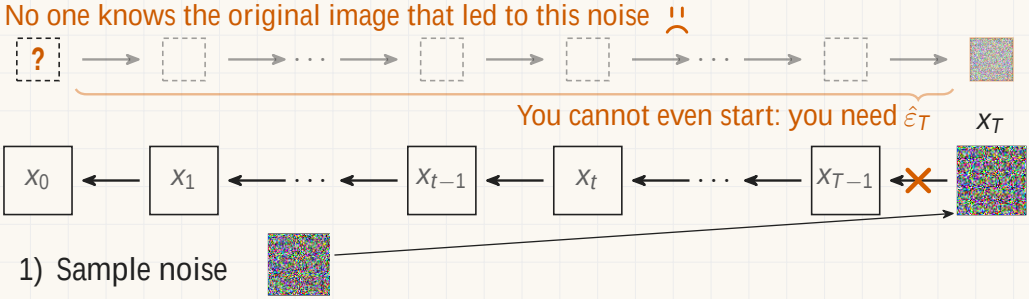
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2) For $t = T, T-1, T-2, \ldots, 1$:

   Estimate x_{t-1} from x_t using $\hat{\varepsilon}_t$

   You don't have it

3) Output x_0

Training of a di□usion model

The model is trained to predict the noise $\hat{\varepsilon}_t$ from the noisy image x_t and the step t.

1. Take an image of the dataset.

x_0



Training of a di□usion model

The model is trained to predict the noise $\hat{\varepsilon}_t$ from the noisy image x_t and the step t.

1. Take an image of the dataset.
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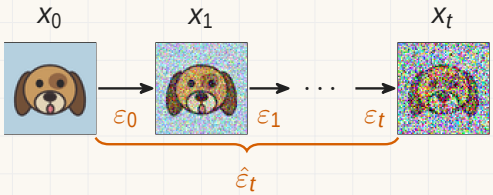
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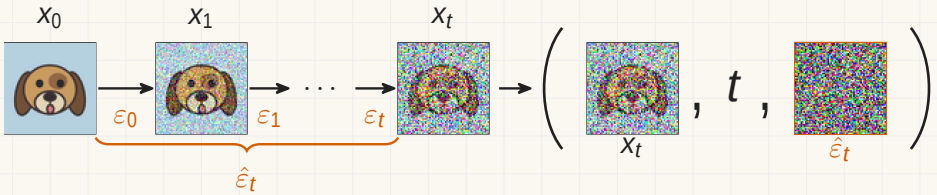
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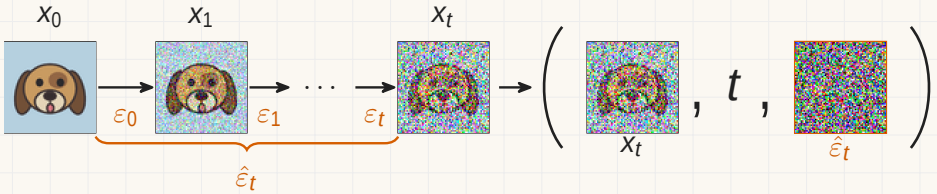
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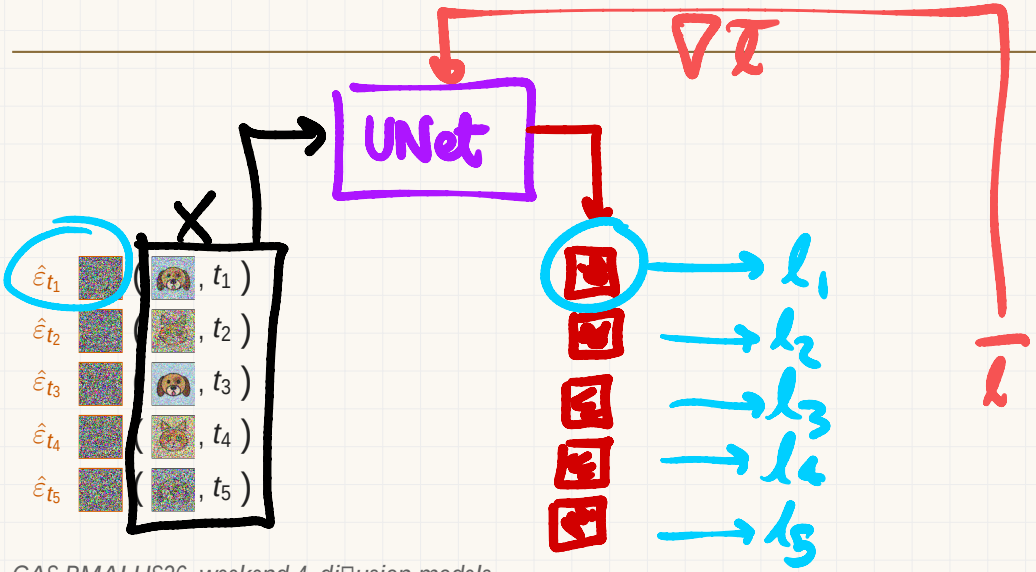
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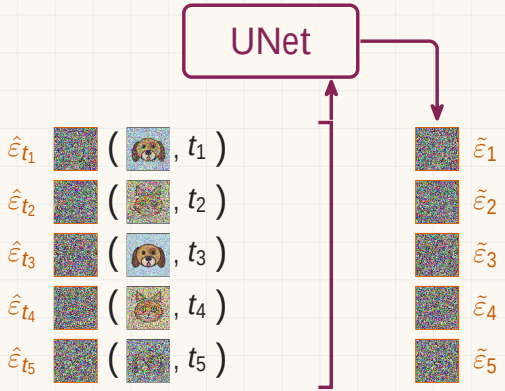


Repeat until you collect $\{(x_t^{(1)}, \hat{\varepsilon}_t^{(1)}), \ldots, (x_t^{(n)}, \hat{\varepsilon}_t^{(n)})\}$

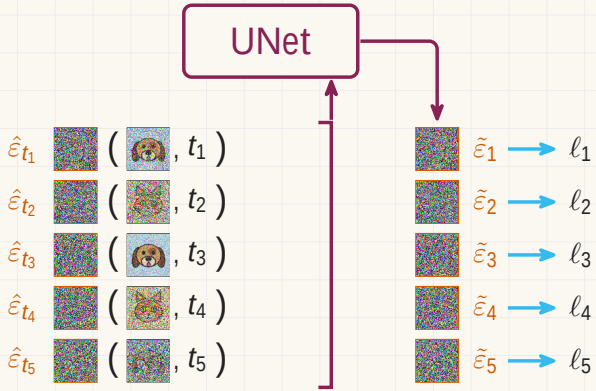
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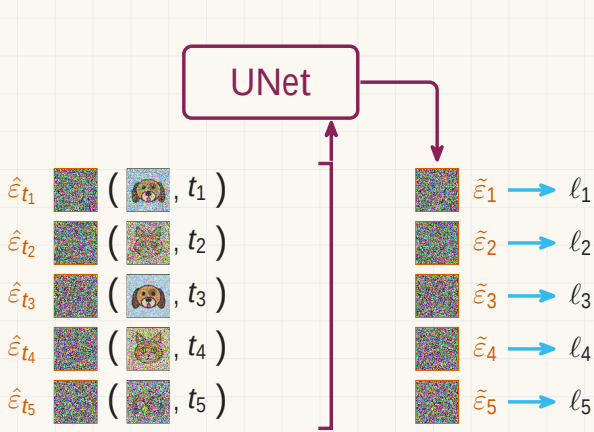
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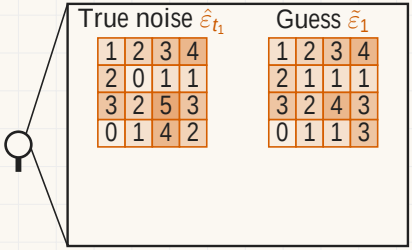


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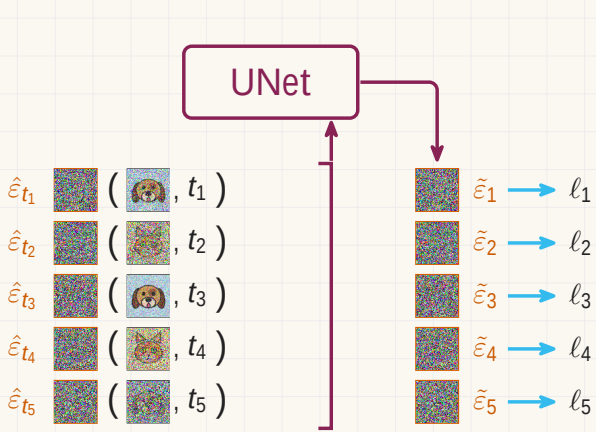


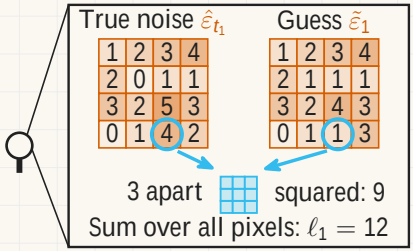
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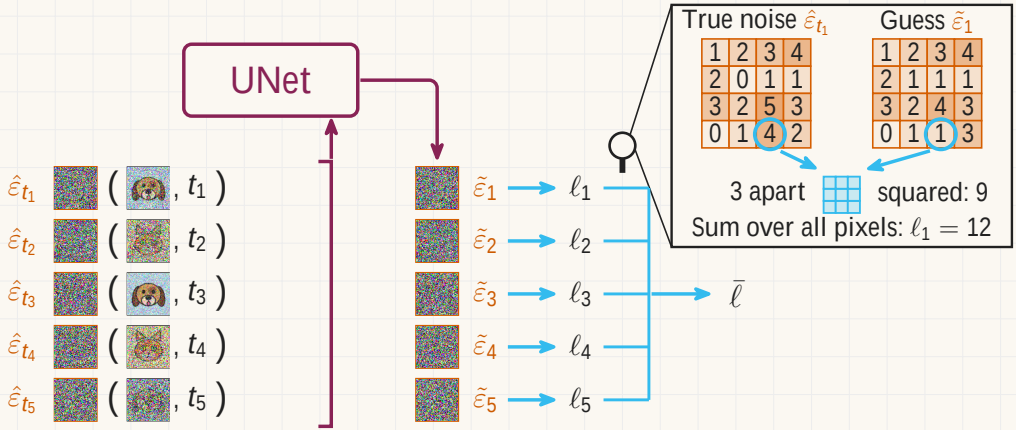


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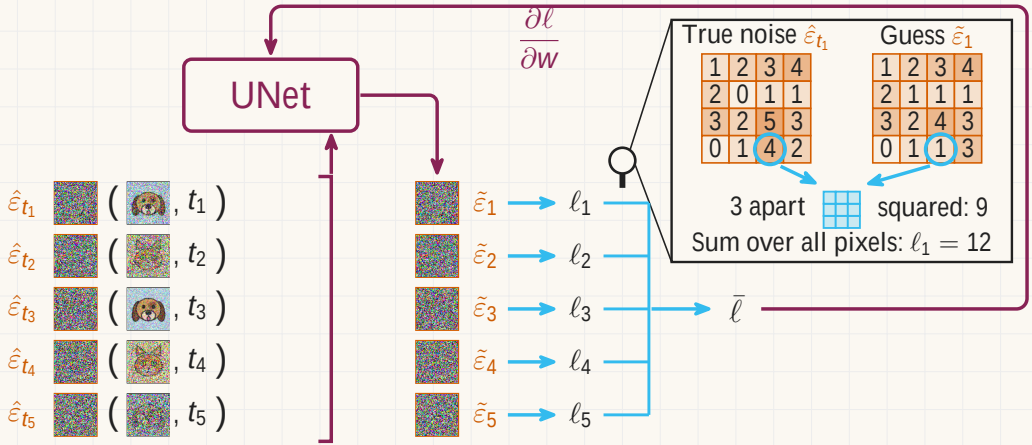




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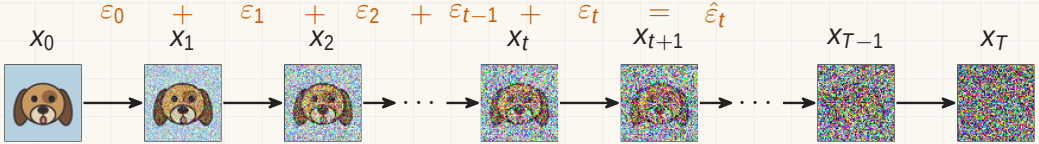
Training a di□usion model



Conditional di□usion models

The prompt "cat" is given to every backward step, so the image it produces is a cat.

Fade & noise

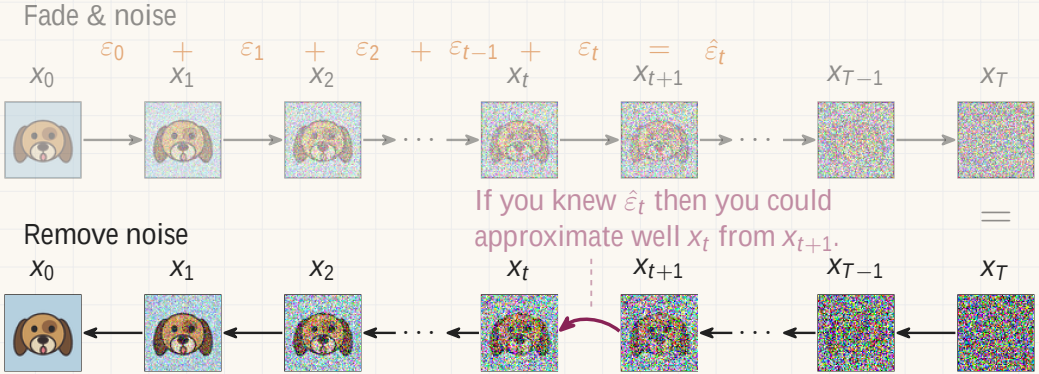


If you knew $\hat{\varepsilon}_t$ then you could approximate well x_t from x_{t+1}.

=

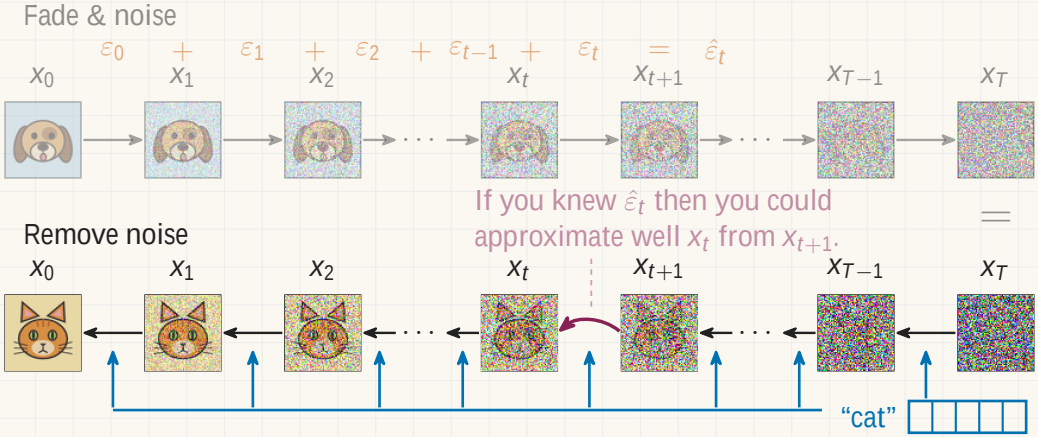
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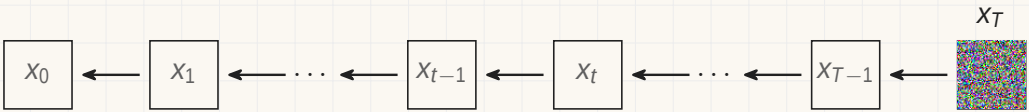
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Generating from di□usion

The prompt "cat" tells every backward step what the image should show.

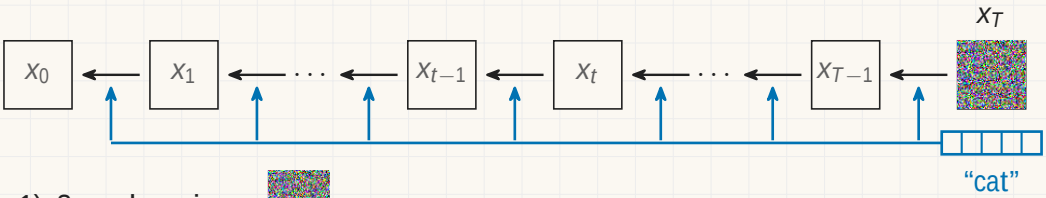


1) Sample noise 
2) For $t = T, T-1, T-2, \ldots, 1$:
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3) Output x_0

You don't have it ☹

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The prompt "cat" tells every backward step what the image should show.



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2) For $t = T, T-1, T-2, \ldots, 1$:

   Estimate x_{t-1} from x_t and "cat" using $\hat{\varepsilon}_t$ ← You don't have it ☹

3) Output x_0

1) Data



"cat"

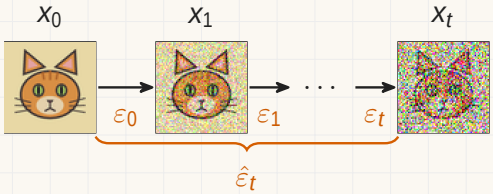


"dog"

Training of a di□usion model

Each training example also carries the caption of its image.

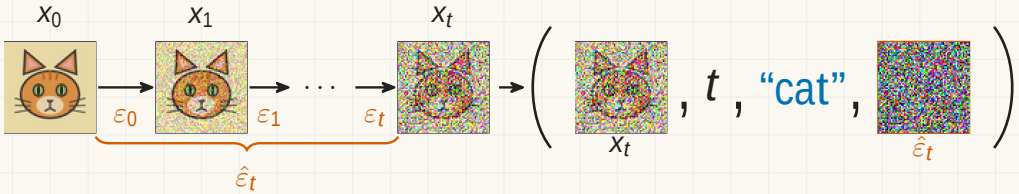
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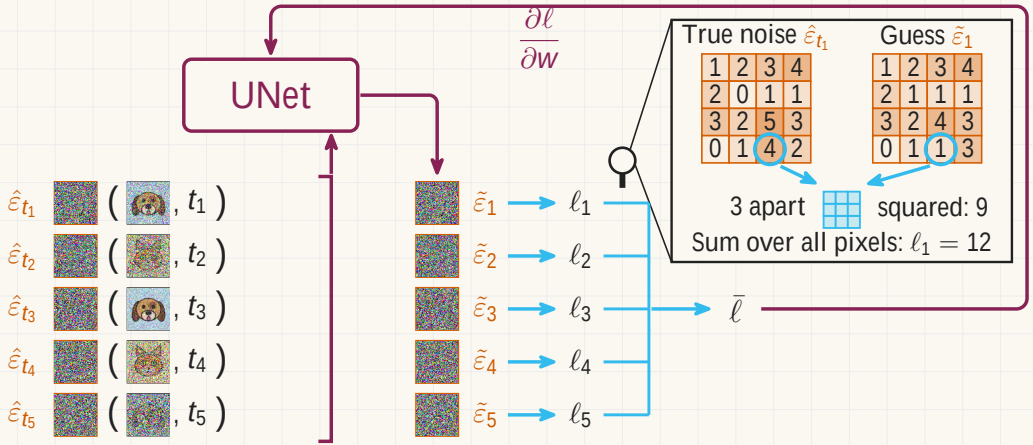
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Training a di□usion model



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